Week 2 Delivery to a target

Instructor practice notes for the revised 15-slide lesson. Connect motors, wheel turns, circumference and both directions of conversion before the robot and cargo-tray rotations. Finish the introductory slides by minute 14; keep the class to one hour.

Class setup

People 12 children in four teams of three; T1/T2 share Kit A and T3/T4 share Kit B. Assign two robot mentors, two build mentors, an adult lead and any floaters. Bring four laptops: two robot-control and two build/reference stations.

Learning goal Each child explains how a motor turns a wheel, predicts distance from wheel degrees, and estimates wheel degrees for a target distance using circumference.

Minimum success Each child predicts or explains both conversion directions, operates a run, and contributes to a tray. Each team records three trials and tests a 40 cm target; a reasoned correction counts even if precision remains imperfect.

Prepare Measure each kit's actual wheel diameter and circumference before class. Check that the drive wheels attach directly to the motor outputs. Mark two clear floor lanes with a 40 cm target measured from one fixed chassis reference. Rehearse 180, 360 and the calculated target value at SPEED = 200. Keep the 0–1080 wheel-degree limit.

Materials Use the complete W02_delivery_Checklist.md and its two-page PDF. Print the 14-page class packet once: five team sheets, four build cards, four wheel reference cards and one attendance sheet. Reuse role cards and hub guides. Bring six 40 cm non-stretch strings or flexible paper strips and calculator access on the laptops. Optional book challenge: two stable books, one per supervised robot station.

Practice Rehearse the motor explanation in two minutes and both conversion examples with a calculator. Test mounted-wheel measurements, the actual 40 cm estimate and tray swaps on both kits. The numerical wheel example is illustrative; it is not a measurement of either classroom kit. Rehearse the optional book challenge only if the table has ample edge clearance; otherwise omit it.

Files Use W02_delivery.pptx with these notes embedded, the full W02_delivery.py and the existing cargo-tray build card. A left/B right, auxiliary motor unplugged, no sensors required. Code uses wheel_degrees; the actual drive helper remains bounded at 0–1080.

Slide 1 Delivery to a target

Timing 0–0:30 minute

Say “Today we will work out how far a wheel should roll, tell the robot what to do, then measure what really happened. Our delivery target is 40 centimeters.”

Connect Show the two assembled bases and a removable cargo tray. Introduce the delivery goal, then use the brief recap on slide 2 to connect it with the first session.

Keep the hour Finish slide 9 by minute 14. The recap replaces part of the welcome. Slides 10–12 are station reference screens, not extra lectures. Each team still has 18 robot minutes. Use the optional challenge only inside an existing retest block after the core trials; stop for reflection at minute 50.

Slide 2 Recap from last session

Timing 0:30–1 minute

Say “Last time, we made the robot show a letter, wait, then move. Let’s read that short recipe together.” Take one or two volunteers; do not turn this into an opening test or ask who remembers nothing.

```
await signal("Y")
await pause(500)
await drive(90)
```

Connect Point to each line as children describe its action. The hub shows Y, waits 500 milliseconds, then asks for 90 wheel degrees. Point to the hub, a motor and its wheel on the stopped base. Today we connect the number in drive to a measured distance.

Interface If useful, point briefly to the recipe and Run/Stop controls in the already-open SPIKE app. Use the large editable recipe on the slide for readability; the student-instructor reference used a whole-screen app image.

Pacing Keep this to 30 seconds. Supply the explanation warmly if no one volunteers. Continue to the motor picture at minute 1.

Slide 3 What makes the robot move

Timing 1–3 minutes

Say “The battery supplies electrical energy. The hub follows our code and controls the motors. Inside a motor, electricity in a coil creates magnetism. Magnetic forces make the motor shaft turn. That turns the wheel.”

Show Point out the real hub, cable, motor and wheel. The picture is a simplified generic motor, not the inside of a LEGO motor. Real motors include extra parts that keep them spinning; do not open the housings or attempt the pictured battery circuit.

Why the whole robot moves “The tire pushes backward on the floor. The floor pushes the tire and robot forward. That grip is friction.” A short mentor-controlled run shows this. If the wheels spin in the air or slip, turning does not guarantee forward distance.

How it knows the amount The SPIKE motor has a built-in rotation sensor. Explain it as a turn counter: it helps the hub know how far the motor output turned. It does not measure the robot's distance along the floor. On this base, the drive wheel is directly attached to that output; added external gearing would change the relationship.

Ask “If a wheel turns while lifted off the floor, will the robot move forward?” No: it needs contact and grip. Do not ask students to hold a running robot.

Sources LEGO medium angular motor: https://education.lego.com/en-us/products/lego-technic-medium-angular-motor/45603/ . Exploratorium motor demonstration: https://annex.exploratorium.edu/xref/exhibits/magnet_electricity_motor.html . Motor artwork is an original generated conceptual illustration; provenance is in 03_Presentations/Assets/W02_Visual_Sources.md.

Slide 4 Degrees measure a wheel turn

Timing 3–4 minutes

Say “Degrees describe part of a turn. A full wheel turn is 360 degrees, half a turn is 180, and a quarter is 90.” Point to the colored sector in each diagram. Have children trace a quarter or half turn in the air with one finger.

Distinguish The command drive(180) asks the paired drive motors for wheel rotation. It does not mean travel 180 centimeters and does not mean turn the robot around. Equal forward wheel rotations make the base go approximately straight. We count one wheel's rotations, not the sum from both wheels.

Ask “What fraction of one turn is 180/360?” One half. “How many turns is 720?” Two. Avoid motor speed here; keep SPEED = 200 throughout the distance test.

Source LEGO New Moves with Motors: https://education.lego.com/en-gb/lessons/get-moving-with-motors/new-moves-with-motors/ . Native editable geometric diagrams show fractions of a turn.

Slide 5 One rotation of the wheel

Timing 4–5 minutes

Say “Watch the dot on the tire. After one full turn it is back at the bottom, but the wheel has moved forward. How could we measure that distance?” Show the rolling example before naming circumference on the next slide. The drawings show the same wheel at two times.

Demonstrate Roll an available loose matching wheel one full turn along a ruler, or use a prepared paper/card circle on the table. If neither is available, trace the diagram and use the measured string. Do not push the assembled robot against its motors to make the demonstration.

Qualify This is a straight rolling prediction without slipping. Real tires can slip, compress or start crooked. Motor rotation feedback counts turns; it does not guarantee exact floor distance. The robot trials will check the prediction.

Ask “What part of the wheel touched the floor during one full turn?” Its whole outside edge. Name that distance around the wheel on the next slide. A larger wheel covers more distance per turn; do not add the left and right wheel distances.

Slide 6 Circumference is the distance around

Timing 5–7 minutes

Say “Diameter goes straight across the center. Radius goes from the center to the edge, so diameter is twice radius: d = 2 × r. Circumference goes all the way around: C = pi × d. Pi is about 3.14.” Connect the circumference to the forward distance in the previous rolling picture. If helpful, substitute d = 2 × r to recover C = 2 × pi × r.

Measure With the robot powered off, show a ruler across the outside of one actual tire. Wrap a non-stretch string or paper strip once around the tire without overlapping, then straighten and measure it. Do not rotate a powered motor by hand or remove a chassis wheel for this activity. A spare matching wheel is useful only if already available.

Example The slide uses an EXAMPLE diameter of 5.6 cm. 3.14 × 5.6 is about 17.6 cm. This is not a claim about the installed wheels: measure both kits and write each kit's C on its team sheets. A different wheel size gives a different distance.

Ask “If we have the radius, what is the diameter?” Twice the radius. “Which measurement tells us the distance for a whole turn?” Circumference. Do not multiply diameter by 360.

Source LEGO Going the Distance discusses circumference and a 5.6 cm example wheel: https://education.lego.com/en-gb/lessons/prime-extra-resources/going-the-distance/ .

Slide 7 Wheel degrees to distance

Timing 7–9 minutes

Say “First find the number of turns: wheel degrees divided by 360. Then multiply by the distance for one turn, our circumference.” Read the formula: predicted distance = (wheel degrees / 360) × C.

Work the example With EXAMPLE C = 17.6 cm, the table shows 360 degrees = one turn = 17.6 cm and 720 degrees = two turns = 35.2 cm. The half-turn answer is deliberately absent. Ask learners to infer 180 degrees before offering the answer: half of 17.6 is 8.8 cm.

Student inference Ask “For YOUR kit's C, how far should 180 degrees move the robot?” They may say half of C before doing any decimal arithmetic. Support this reasoning even if a mentor handles the calculator.

Units Use centimeters for both C and the predicted distance. The formula predicts distance; increasing speed is not the same as increasing the number of wheel turns. Keep speed unchanged in this lesson.

Slide 8 Target distance to wheel degrees

Timing 9–11 minutes

Say “Work backward from where we want to go. Target distance divided by circumference gives the turns we need. Multiply those turns by 360 to get wheel degrees for the code.”

Work the example For EXAMPLE C = 17.6 cm and a 40 cm target: 40 cm / 17.6 cm is about 2.27 rotations. Calculate (40 / 17.6) × 360 without rounding the intermediate result; round only the final code value to 818 wheel degrees. The slide labels rotations and degrees once each. Use our actual kit circumference for a real run.

Both directions Check by substituting: (818/360) × 17.6 is about 40 cm. For the actual robot, use its own C. More than one answer can be reasonable after calibration; the formula gives an estimate that needs a test.

Meters question If the request is 1 meter, convert it to 100 cm before dividing by C. For example C = 17.6 cm, that would need about 2045 degrees. This is a PAPER calculation only: it exceeds today's 1080-degree helper limit and lane. Do not remove the guard or run this extension.

Prediction prompt “What number should replace 180 to aim for 40 cm on OUR kit, and why?” Expected explanation: divide 40 by measured C, multiply by 360 and round once at the end.

Slide 9 Tell the code how far the wheels turn

Timing 11–14 minutes

```
wheel_degrees = 180
await drive(wheel_degrees)
await signal("D")
```

Point wheel_degrees = 180 stores a wheel-rotation amount; that assignment alone does not move anything. await drive(wheel_degrees) asks the prepared helper to turn the two drive wheels that amount. await signal("D") displays D after the move. D is not a sensor check that the target was reached.

Clarify the old wording C means the measured wheel circumference. Keep the distance formula visible beside the recipe: predicted cm = (wheel degrees / 360) × C. Ask explicitly for centimeters at 180 and 360 degrees, then ask which wheel_degrees value should aim for 40 cm. The variable stores wheel rotation, not centimeters.

Live check Run one short rehearsed 180-degree demonstration at the marked start and measure the fixed chassis reference. Ask a child to point to the number they will change. Keep names matching in assignment and call. Paste the full W02_delivery.py; the slide is only its student recipe.

One change Hold the same chassis reference, floor, kit, heading, speed, mentor tray and cube fixed for the three calibration runs. Change only wheel_degrees. Save the team tray for a separately labeled attachment test.

If behind Do not add more lecture after minute 14. Use the wheel reference card and calculator while coaching at the robot. The key reasoning is fractions of a turn and the reverse question, not memorizing pi.

Slide 10 Our robot and build jobs

Timing 14–38 minutes, then retests

Station map 14–26: T1/T3 use robots; T2/T4 build. 26–38: switch. 38–44: T1/T3 retest; 44–50: T2/T4 retest. Rotate jobs at 18, 22, 30, 34, 40, 42, 46 and 48. Transitions fit inside the blocks; each team has 18 robot minutes.

Robot jobs Driver edits and starts after “clear.” Navigator predicts, places the stopped robot/cube and checks the lane. Observer measures the same chassis mark and records centimeters. Each child drives one of the three initial trials; all three rotate and explain.

First turn Align a fixed mark on the chassis with the start line. Use measured circumference C to predict (180/360) × C cm. Run wheel_degrees = 180, then measure displacement of that same mark. Keep the mentor tray and cube fixed. Swap driver. Predict (360/360) × C cm, then run wheel_degrees = 360. Compare the measured distance with C. Keep the same kit, floor, start direction, speed, tray and cube. Two driven wheels do not double the distance. The third driver enters round((40/C) × 360) as wheel_degrees and tests the 40 cm target. Use a calculator and the kit's own measured C. Example only: C = 17.6 cm gives about 818 degrees. Do not exceed 1080; ask the mentor to check measurements if the answer is outside the allowed range.

Build activity Use slide 12 and the build card to make a removable cargo tray/cart that holds the cube without hindering robot movement. Check the mounting proxy first, then fit to a stopped robot with the mentor. Keep the chassis and drive wheels assembled.

Retest Repeat or adjust the 40 cm setting with the SAME mentor tray and cube first. Then test the team cart at the same code setting, labeled as a separate attachment test. If this is complete and time remains, use slide 14 within the same retest block; do not extend the hour.

Build connection Inspect clearance before mounting a stopped robot. Keep wheels, controls and cables clear. Store four labeled trays. A useful tray keeps the cube aboard without sagging into a wheel; it is not a reason to rebuild the drivetrain.

Slide 11 Programming exercise: reach 40 cm

Timing Station reference during 14–38; corrections during 38–50

Goal Use wheel rotation to predict and test travel. Keep the same kit, SPEED = 200, heading, floor, mentor tray and cube for the main comparison. Only wheel_degrees changes.

Three drivers Driver 1 predicts and tests 180 degrees. Driver 2 predicts and tests 360. Driver 3 calculates round((40 / C) × 360), predicts the result and tests the 40 cm target. Use measured kit C and a calculator; keep the helper limit 0–1080.

Measure Align the same fixed chassis mark with the start each time, then measure its displacement in centimeters. Do not switch to the front of the cart in these calibration trials. The observer records degrees, predicted cm and measured cm on the existing team sheet.

Explain Before pressing Run, the navigator explains why the next number should work. The driver edits only the student recipe and starts after the mentor says clear. Swap jobs every four minutes in the first turn.

Improve During retest, say how many centimeters short or far the robot stopped, propose one change and explain it. Repeat a suspect measurement before changing the estimate. The motor counts rotation; it does not see the target.

Build connection After calibration, attach the team cart while stopped and repeat the same code with the same cube. Label this an attachment test, so changes to the build are not confused with changes to the program.

Slide 12 Build exercise: a cart that moves freely

Timing Station reference during 14–38; fit and movement check during retest

Goal Build a removable cart/tray that keeps the cube aboard and does not hinder the robot's movement. This is the existing cargo-tray activity: it does not require a new wheeled chassis or powered attachment.

Build Make a flat support, add a low edge or rail, and attach it securely using the approved mount. Use beams/pins or pre-cut card/tape. Use the mounting and clearance template while the robot is at the other station.

Check movement The cart must not rub the drive wheels, drag on the floor, snag cables or block the hub controls. Keep the load light, balanced and secure. A cart can hold its cube yet still be a poor design if it slows or blocks the robot.

Hand test With the cart off the robot, tilt it gently. Observe whether the cube stays aboard; change one rail, support or attachment detail and compare. Builder connects, designer checks fit and parts, tester loads and reports. Rotate every four minutes.

Fit and test A mentor checks clearance and fits the cart to the stopped robot. Use a short rehearsed run in a clear lane with the same code and payload. Stop if the cart drags, rubs or shifts; fix the build before changing the program. Do not push the assembled drivetrain by hand.

Evidence Record one build change and what improved. Store four labeled team carts. If there is time after the core robot trials and this clearance check, the combined book-parking challenge uses the attached cart.

Slide 13 Compare prediction with measurement

Timing 38–50 minutes within the retest blocks

Evidence Record wheel degrees, predicted centimeters and measured centimeters for three initial runs. Use the margin or back for retests. For the target trial, say how many centimeters too short or too far. Keep the fixed chassis mark as the measurement point even when the tray changes.

Ask “How close was our prediction? What could explain a miss? What one change should we try next?” Listen for wheel size, slip, start alignment or payload; not “the robot knew where 40 cm was.” The program does not sense the target.

Calibration If a careful 360-degree run differs from geometric C, repeat that run before adopting its measured distance as an effective cm-per-turn value. Then estimate 360 × target / measured_cm_per_turn. Keep physical circumference and measured travel labeled separately; they need not match exactly.

Support Use half, one and two turns; let a mentor enter the calculator steps. Do not require everyone to finish independent long division. A reasoned correction is useful evidence even if the robot does not land within 5 cm.

Stretch Predict a different target within the rehearsed lane and helper limit, or repeat one setting to see variation. Explain the 1 meter calculation on paper only. Do not change speed during the main distance comparison.

Fallback If Python cannot be restored quickly, use a rehearsed Word Blocks sequence with 0.5 or 1 motor rotations. One rotation is 360 degrees. The word-block speed percentage is not the Python velocity value; do not copy speed numbers between them.

Slide 14 Optional team challenge: park by a book

Timing Only within a team's 38–44 or 44–50 retest block after core work

Set up Builders attach the checked cart to the stopped robot. The instructor chooses a new starting spot on the table and places a stable book straight ahead. Align the robot with the book; choose a short clear route well inside every table edge. Supervise throughout with Stop ready. Skip the tabletop run if there is not ample space for the full robot, cart and stopping margin.

Team problem Ask: "What must change in the code to park as close as we can without touching the book?" Let the team decide how to measure and calculate before showing the calculation. No distance sensor is used: the robot cannot detect the book or a table edge.

Measure Measure the gap in centimeters from the FRONTMOST part of the robot/cart/load to the near face of the book. Agree a visible clearance, initially about 3 cm. Planned travel = measured gap − chosen clearance. If that is zero or negative, reposition; do not run.

Calculate Wheel degrees = (planned travel / measured C) × 360. Example only: gap 30 cm, clearance 3 cm, C = 17.6 cm gives (27 / 17.6) × 360 ≈ 552 wheel degrees. The programmer replaces the value of wheel_degrees in the existing recipe; keep SPEED = 200 and the 0–1080 limit. Shorten the route if it exceeds the limit.

Test and refine The mentor checks the estimate and clear path before one supervised run. Measure the remaining gap. If time permits, reset to exactly the same marked start with the same cart, cube and book; explain one code adjustment and retest. No part may touch the book. Do not pursue smaller gaps when control or stopping space is uncertain.

Roles and record Builders check fit and front clearance; the team reasons through the measurement; the programmer edits; the observer records the result on the back of the team sheet. Leave the answer to the team before coaching. The closest non-contact result is useful evidence, not a reason to rush or skip core work.

Timing Stop at the end of that team's existing retest block. Reflection still starts at minute 50 and save/reset at 54. This optional challenge adds no required printouts or sensor lesson.

Slide 15 Explain the motion

Timing 50–54 minutes reflection; 54–60 save and reset

Ask each child Choose one: “What makes the motor turn?” “How far should half a wheel turn go?” “How would you choose the code number for a target?” If the optional challenge was tried, ask how the cart’s front and a clearance gap changed the calculation. Give each child a brief speaking turn in the two kit groups.

Listen for Electrical energy and magnetic forces turn the motor; wheel grip moves the robot. Half a turn predicts half the circumference. For a target, divide its distance by C then multiply by 360, keeping units alike. Actual travel still needs measuring.

Close Save T1–T4 projects and backups. Stop hubs, remove only approved attachments, label/store four trays and count parts. Collect records privately. Follow YMCA pickup procedures and assign charging.

After class Record actual wheel measurements, rehearsal problems and observed learning in SESSION_PREP.md or the approved private learner records. The November 7 report confirms a Session 2 practice and requested revisions, not a completed second class or all hardware checks.